

# Software Design Diagram Generator

## Master Sprint Plan & EPIC Documentation v2.0

Course: IT314 Software Engineering

Group: 27

Methodology: Agile (Scrum)

155 Requirements | 13 EPICs | 4 Sprints

155

USER STORIES

13

EPICS (E-A TO E-M)

4

SPRINTS (2-WK)

6

UML & DFD TYPES

### 1. EPIC MASTER OVERVIEW

HIGH-LEVEL ROADMAP

| EPIC | NAME & SCOPE                             | SRS REF             | COUNT | SPRINTS (S1/S2/S3/S4) | DEPENDS ON    |
|------|------------------------------------------|---------------------|-------|-----------------------|---------------|
| E-A  | Accounts, Auth & Security Foundation     | FR28, NFR3-5        | 13    | 7 / 4 / 2 / 0         | E-M           |
| E-B  | Project & Document Ingestion             | FR1, FR2, FR3       | 17    | 3 / 11 / 3 / 0        | E-A, E-M      |
| E-C  | Requirement Extraction & Validation      | FR4, FR5, FR6-8     | 23    | 10 / 8 / 5 / 0        | E-B           |
| E-D  | Diagram Generation                       | FR9-11, FR30, DR1-2 | 25    | 6 / 8 / 7 / 4         | E-C           |
| E-E  | Diagram Review, Editing & Versioning     | FR12-15, FR29       | 19    | 0 / 6 / 8 / 5         | E-D           |
| E-F  | Traceability & Cross-Diagram Consistency | FR16, FR17          | 7     | 1 / 1 / 4 / 1         | E-C, E-D      |
| E-G  | Export & Documentation                   | FR18                | 6     | 1 / 2 / 2 / 1         | E-D           |
| E-H  | Onboarding, Guidance & Feedback          | FR19, FR27          | 5     | 0 / 0 / 0 / 5         | E-C, E-D, E-J |
| E-I  | PM Reporting                             | FR20, FR21, FR22    | 7     | 0 / 0 / 2 / 5         | E-C, E-D      |
| E-J  | Admin & Collaboration                    | FR23, FR24, FR25    | 15    | 0 / 0 / 7 / 8         | E-A, E-B      |
| E-K  | Search & Navigation                      | FR26                | 4     | 0 / 0 / 0 / 4         | E-B, E-C, E-D |
| E-L  | Platform Quality & Compatibility         | NFR1-2, NFR6-9      | 8     | 0 / 1 / 1 / 6         | Cross-cutting |
| E-M  | Project Setup & Technical Enablers       | TE1 - TE6           | 6     | 5 / 0 / 1 / 0         | None          |

### 2. SPRINT RELEASE STRATEGY

STORY DISTRIBUTION: 33 / 41 / 42 / 39

| SPRINT   | STRATEGIC FOCUS & DELIVERY GOAL                                                                                                                                                                               | STORY COUNT |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Sprint 1 | <b>Walking Skeleton:</b> Email auth, project creation, paste input, extract items, Use Case diagram generation, PNG export, Vercel/Render deployment, concept poster.                                         | 33 Stories  |
| Sprint 2 | <b>Ingestion &amp; Validation:</b> PDF/DOCX multi-upload, input checks, requirement flagging, Class & Activity diagrams, inline diagram editing, SVG/PDF export, Google OAuth, encryption at rest.            | 41 Stories  |
| Sprint 3 | <b>Complete Generation &amp; Versioning:</b> Sequence & DFD diagrams, flag resolution ("Generate anyway"), requirement change propagation, diagram auto-validation, PlantUML/XML export, collaboration roles. | 42 Stories  |
| Sprint 4 | <b>Harden &amp; Customise:</b> Component diagrams, partial regeneration, email alerts, custom styles, PM reporting PDF, search/navigation, admin dashboard, quality target verifications.                     | 39 Stories  |

### 3. DETAILED BACKLOG – SPRINT 1

33 STORIES | WALKING SKELETON

#### Sprint 1 Objective

Deploy an end-to-end walking skeleton to Vercel/Render. Enable users to register, create a project, paste raw requirement text, extract domain elements, confirm entities, generate a standard UML Use Case diagram, and export it to PNG.

| EPIC | STORY ID           | REQUIREMENT / GOAL STATEMENT                                                                                                         |
|------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| E-A  | FR28a, c, d        | Sign up with email/password, log in with credentials, log out securely.                                                              |
| E-A  | NFR3b, 4a, 5a, 5d  | In-transit TLS encryption, login enforcement, view privacy policy, AI consent prompt.                                                |
| E-B  | FR1a, FR3a, FR3c   | Type/paste requirement text directly, create named project, view personal project list.                                              |
| E-C  | FR4a - FR4g        | Split text into requirement IDs; extract actors, classes, processes, data flows, interactions, & relationships.                      |
| E-C  | FR5a, FR5b, FR5e   | View consolidated extracted data in pre-gen dashboard, edit extracted items, explicitly confirm mapping.                             |
| E-D  | FR9d, FR10a, FR10g | Confirm diagram selection, generate standard Use Case diagram, display live progress spinner.                                        |
| E-D  | FR30c, DR1a, DR2a  | Server wake-up toast, strictly follow Use Case UML standard, flag unbacked items as "assumption".                                    |
| E-F  | FR16a              | Generate all diagrams from a single, shared underlying project requirements model.                                                   |
| E-G  | FR18a              | Export generated Use Case diagram as high-resolution PNG image.                                                                      |
| E-M  | TE1 - TE5          | GitHub repository with CI actions, automated Render/Vercel deployment, DB/storage, AI proxy module, transactional email integration. |

### 4. DETAILED BACKLOG – SPRINT 2

41 STORIES | INGESTION & FIRST VALIDATION

#### Sprint 2 Objective

Expand input options to PDF/DOCX files with guardrails (10MB / 100 pages), introduce initial quality checks (ambiguity/contradiction flagging), generate Class & Activity diagrams, support inline graphical edits, SVG/PDF export, Google sign-in, and per-project permissions.

| EPIC | STORY ID            | REQUIREMENT / GOAL STATEMENT                                                                                                                  |
|------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| E-A  | FR28b, e, NFR3a, 4b | Google OAuth login, password reset flow, AES-256 data encryption at rest, per-project authorization.                                          |
| E-B  | FR1b, c, FR2a-c     | Upload PDF SRS, upload DOCX SRS, warn on >10MB files, reject corrupted/unsupported formats, alert on non-extractable PDF text.                |
| E-B  | FR3b, d, e, f, h, i | Rename project, switch active projects, delete project, upload multiple docs per project, enforce limit of 15 docs and 100 pages per project. |
| E-C  | FR5c, d, FR6a-d     | Add missing elements manually, delete incorrect items, auto-flag ambiguous, incomplete, inconsistent, and contradictory requirements.         |
| E-C  | FR7a, FR7b          | Visually highlight flagged requirements in editor, display rationale explanation for each flag.                                               |
| E-D  | FR9a, 10b, c, 11a   | Display AI diagram recommendations with reasons, generate Class diagrams, generate Activity diagrams, show in-app completion notification.    |
| E-D  | FR30a, b, DR1b, c   | Auto-retry on AI rate limits, clear error prompt with retry on failure, enforce standard UML for Class and Activity diagrams.                 |
| E-E  | FR12a - FR12f       | Add element to canvas, remove element, rename element, reposition/drag element, change relationship types, edit multiplicity & stereotypes.   |

## 5. DETAILED BACKLOG – SPRINT 3

42 STORIES | GENERATION & CHANGE MGMT

| EPIC | STORY ID              | REQUIREMENT / GOAL STATEMENT                                                                                                                                                                 |
|------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-A  | NFR5b, NFR5c          | Auto-delete inactive account data after 30 days, support immediate user data wipe request.                                                                                                   |
| E-B  | FR1d, e, FR3g         | Input user stories directly, input FR/NFR tables, mix text + PDF + user story inputs in 1 project.                                                                                           |
| E-C  | FR7c - e, FR8a - b    | AI clarification questions for flags, manual flag resolution, AI fixes, "Generate Anyway" bypass.                                                                                            |
| E-D  | FR9b, c, 10d, e       | Add non-recommended diagram types, remove unwanted types, generate Sequence diagrams, generate DFDs.                                                                                         |
| E-D  | DR1d, e, DR2b         | Enforce Sequence UML standards, enforce DFD standards, distinct visual styling for assumption nodes.                                                                                         |
| E-E  | FR13b, 14a-e, 29a, b  | Regenerate full diagram, modify source text, highlight affected diagrams, auto-regenerate on approval, diagram version history, revert to prior version, diagram syntax validation warnings. |
| E-F  | FR16b, FR17b - d      | Propagate shared model updates, list unrepresented requirements, prompt update on change, drift warnings.                                                                                    |
| E-G  | FR18d, FR18e          | Export diagrams in editable draw.io XML format and PlantUML raw code representation.                                                                                                         |
| E-I  | FR22a, FR22b          | Audit logging for requirement edits and diagram generation/edit actions.                                                                                                                     |
| E-J  | FR23a-c, 24a, b, e, f | User account listing, create user, deactivate user, invite project collaborators, assign Viewer/Editor/Owner roles, enforce role RBAC, simultaneous edit collision warnings.                 |
| E-L  | NFR1a                 | Target generation speed: Complete diagram synthesis within NFR1 limits for benchmark docs.                                                                                                   |
| E-M  | TE6                   | Implement background scheduled-job runner for cleanup, audit maintenance, and async tasks.                                                                                                   |

## 6. DETAILED BACKLOG – SPRINT 4

39 STORIES | FINISH, HARDENING & QUALITY TARGET VERIFICATION

| EPIC | STORY ID               | REQUIREMENT / GOAL STATEMENT                                                                                                                                                                 |
|------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-D  | FR10f, 11b, c, DR1f    | Generate Component diagram, email notifications on completion, toggle email alerts, Component UML standards.                                                                                 |
| E-E  | FR13a, 15a-c, 29c      | Sub-diagram partial selection regeneration, change color themes, change layout orientation (LR/TB), adjust canvas font size, explicit "Accept diagram validation as-is" override.            |
| E-F  | FR17e                  | Cross-diagram naming inconsistency alerts (e.g., entity named "Client" vs "Customer").                                                                                                       |
| E-G  | FR18f                  | One-click fast export button using primary preferred format defaults.                                                                                                                        |
| E-H  | FR19a-b, FR27a-c       | Interactive pipeline step-by-step onboarding walkthrough, contextual UML notation guide tooltips, report diagram bug, submit requirement feedback, view submitted feedback reports.          |
| E-I  | FR20a-c, FR21a-b       | PM Dashboard metrics (diagram coverage %, ambiguity count, unrepresented count), compile executive summary report, download consolidated PDF documentation package.                          |
| E-J  | FR23d, 24g, 25a-f      | Reactivate user, remove collaborator, admin analytics dashboard (active users, total projects, storage vs quota, error logs, AI generation latencies, view active app configs).              |
| E-K  | FR26a - d              | Search requirements and diagrams by keyword, search by ID, filter by status, filter by diagram type.                                                                                         |
| E-L  | NFR2a, 6a-b, 7b, 8, 9a | 1-hr end-to-end usability benchmark, incremental regeneration performance, maintenance schedule banner, verified multi-browser support (Chrome, Firefox, Edge), 5-user concurrent load test. |

## 7. CROSS-SPRINT ARCHITECTURAL DECISIONS

EVOLUTION STRATEGY

### 1. Requirement Flagging vs. Generation Blocking

Requirement quality flags (ambiguity, contradictions) are detected from **Sprint 2** (FR6, FR7a-b). Until explicit flag resolution ("Generate anyway", FR8a) arrives in **Sprint 3**, detected flags will remain prominently visible in the editor but will *not hard-block* diagram generation.

### 2. Diagram Selection Recommendations

In Sprint 1, the diagram choice is fixed to standard Use Case diagrams. Diagram type recommendation logic (FR9a) ships in **Sprint 2**, and manual adding/removing of recommended types (FR9b-c) arrives in **Sprint 3**.

### 3. Versioning & History Tracking

Every user save action creates an immutable database version snapshot starting in **Sprint 2** (FR12). However, the UI history viewer and point-in-time rollback interface (FR14d, FR14e) ship in **Sprint 3**.

### 4. Multi-User Project Sharing

The dashboard project list shows strictly self-owned projects in Sprints 1 and 2. The "Shared with Me" project tab and granular RBAC roles (FR24a-f) are activated when collaboration features land in **Sprint 3**.

## 8. SPRINT 1 WORKSTREAM ALLOCATION & PLANNING POKER

TEAM OF 10 (5 PAIRS)

**Workstream Legend:** FE Frontend | BE Backend & Data | AI Requirement Parsing & Extraction | DG Diagram Generation & Rendering | OPS DevOps, Security & QA

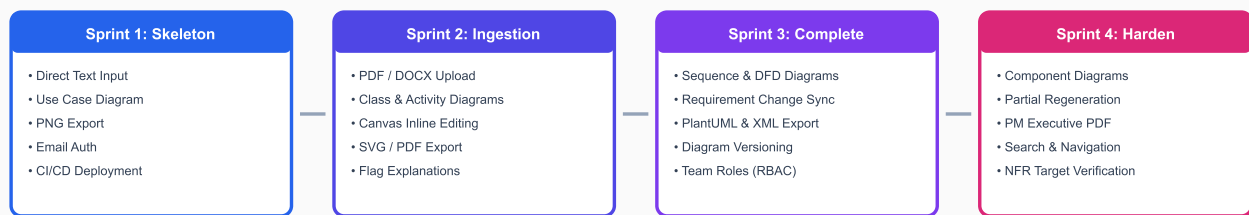
| ID          | STORY GOAL                                                                | STREAM | ASSIGNEE     | POINTS |
|-------------|---------------------------------------------------------------------------|--------|--------------|--------|
| FR1a        | Type/paste requirement text directly                                      | FE     | Pair 1 (FE)  | 2 pts  |
| FR3a, c     | Create named project & view project list                                  | FE/BE  | Pair 2 (BE)  | 3 pts  |
| FR4a-g      | Parse text & extract actors, entities, processes, data flows              | AI     | Pair 3 (AI)  | 8 pts  |
| FR5a, b, e  | View extracted info, edit extracted items, confirm mapping                | FE/BE  | Pair 1 (FE)  | 5 pts  |
| FR9d, 10a   | Confirm selection & generate Use Case diagram                             | DG     | Pair 4 (DG)  | 5 pts  |
| FR10g, 30c  | Progress spinner & server wake-up notification                            | FE     | Pair 1 (FE)  | 2 pts  |
| FR16a       | Shared requirement model data structure                                   | BE     | Pair 2 (BE)  | 3 pts  |
| FR18a       | Export diagram to PNG format                                              | DG     | Pair 4 (DG)  | 3 pts  |
| FR28a, c, d | Email/password sign-up, login, and logout                                 | BE     | Pair 2 (BE)  | 5 pts  |
| NFR3b, 4a   | HTTPS transit encryption & login authentication guard                     | OPS    | Pair 5 (OPS) | 2 pts  |
| NFR5a, 5d   | Display privacy policy & AI consent modal                                 | FE     | Pair 1 (FE)  | 2 pts  |
| DR1a, 2a    | Standard Use Case UML rules & visual assumption tagging                   | DG     | Pair 4 (DG)  | 3 pts  |
| TE1 - TE5   | GitHub CI, Vercel/Render pipeline, DB setup, AI module proxy, Email setup | OPS/BE | Pair 5 + 2   | 13 pts |

Total Sprint 1 Story Points: 56 Points

Figure 1: End-to-End Requirement Processing Pipeline



Figure 2: Sprint Execution Progression & Capability Delivery



## 10. WORKING AGREEMENTS

|                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition of Ready (DoR)</b> <ul style="list-style-type: none"><li>Acceptance criteria (✓ pass and ✗ fail cases) defined in backlog.</li><li>Story estimate agreed upon by team via Planning Poker.</li><li>Prerequisites done or scheduled in same sprint.</li><li>Fits within single 2-week sprint capacity; else split.</li></ul> | <b>Definition of Done (DoD)</b> <ul style="list-style-type: none"><li>All acceptance criteria pass automated/manual testing.</li><li>Merged to main via reviewed PR; CI pipeline green.</li><li>Verified working on live environment (Vercel + Render).</li><li>NFR1 generation timing benchmark verified; docs updated.</li></ul> |
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## 11. RISK MATRIX & MITIGATION STRATEGY

| #  | IDENTIFIED RISK EVENT                                                                | MITIGATION & CONTINGENCY PLAN                                                                                                                                  | MAPPED MODULES    |
|----|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| R1 | AI provider free-tier rate limits or API quota exhaustion halts diagram generation.  | Implement async queueing with exponential backoff retries (FR30a); isolate AI layer in module proxy (TE4); surface live failures in admin dashboard (FR25d/e). | FR30, TE4, NFR9a  |
| R2 | Free web hosting tier (Render) enters sleep state, causing first request timeout.    | Deploy 10-min automated ping uptime monitor; display clear "Server is starting..." wake-up notification state in frontend UI (FR30c).                          | TE2, FR30c, NFR7a |
| R3 | External AI provider terms permit training on user submitted SRS data.               | Mandatory user AI consent prompt (NFR5d) and explicit Privacy Policy disclosure (NFR5a); enforce zero-data-retention provider contracts.                       | NFR5a, NFR5d      |
| R4 | Sprints 2 and 3 carry high story counts (41 & 42 stories), risking schedule overrun. | Re-evaluate team velocity post-Sprint 1; strictly move non-critical stories along dependency chains; preserve Sprint 4 buffer.                                 | Section 2 & 6     |
| R5 | Free email sending provider rate limits block account verification emails.           | Integrate single transactional provider; log failed deliveries and retry once; set default email notification settings to OFF.                                 | TE5, FR11         |
| R6 | Initial user requirement elicitation sample size is thin or incomplete.              | Enforce close-out acceptance criteria in Lab 6 documentation; convert pending user ambiguities into formal backlog refinement spikes.                          | Lab 6 Document    |

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Final